

Vian Garg

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Education

Carnegie Mellon University

Bachelor of Science in Electrical and Computer Engineering, Minor in Robotics
GPA: 4.0/4.0 (Dean's List)

May 2029
Pittsburgh, PA

Work Experience

Robotics Researcher

Engineering Materials for Transformative Technologies Lab

May 2026 – Present

Pittsburgh, PA

- Developing virtual cobot welding system bridging human motion input to UR10e execution via URScript.
- Built Python pipeline converting CV-detected weld waypoints into cobot-executable URScript instructions.
- Engineered remote validation infrastructure for script upload and testing over CMU VPN via SSH.

Aerodynamics Engineering Intern

HALCON, EDGE Group

June 2024 – August 2024

Abu Dhabi, UAE

- Designed a split-canard rocket in OpenRocket and simulated full flight trajectories.
- Performed CFD and stability analysis using Ansys Fluent, XFLR5, and modular rocket models.
- Built state-space tools: coded matrix calculator, optimized stability gains, and created PyGame flight sim.
- Developed OOP + ML controller using NEAT algorithm, achieving a 93% targeting hit-rate.

Project Experience

NASA Lunabotics Robotics Engineer

- Developed high-payload autonomous lunar rover; won Caterpillar First Steps Award as best first-year team.
- Designed and fabricated metal track system achieving a drawbar pull ratio of 1.55 on BP-1 simulant.
- First first-year team to score autonomous navigation points at NASA Lunabotics competition.
- Tuned Nav2 voxel costmap parameters to match robot's physical mobility constraints and track geometry.

January 2026 – Present

SPIRIT Buggy Assistant Build Lead

- Led build of SPIRIT's first new CF buggy design in 15+ years, cutting race time by 5 seconds (3%).
- Introduced the first male-mold construction ever used in CMU Buggy, improving surface finish and aerodynamics.
- Incorporated new-gen Nomex core and CF weave layup for structural efficiency and mass reduction.
- Designed steering assembly in SolidWorks; validated structural margins under load with ANSYS.

August 2025 – Present

Flight Simulator — Full Stack

- Coded physics-based flight simulator with 6DOF mechanics and 5D state-space dynamics without libraries.
- Awarded \$1000 AppLovin prize at TartanHacks, competing against 1200+ hackathon competitors.
- Integrated Ursina for 3D visualization, cutting load time by 8.37s and improving responsiveness.
- Optimized for low-spec hardware (4GB RAM) to support flight mechanics education.

December 2024 – January 2025

Leadership Experience

Rethink the Rink Engineer

CMU Manufacturing Futures Institute | Covestro, Pittsburgh Penguins, Bauer Hockey

January 2026 – March 2026

Pittsburgh, PA

- Engineered goalie chest protector addressing impact absorption, mobility, and thermal management.
- Developed hexagonal impact plate system with cooling mechanism inspired by motorsport applications.
- Designs licensed by Covestro; selected for potential commercialization with Bauer Hockey.

National Robotics Vice President

FIRST Global Team UAE

January 2024 – October 2024

Athens, Greece

- Led 22-member national robotics team in world's largest international robotics competition.
- Engineered record-breaking robot with 6ft telescoping cascade lift and omni-wheel drivetrain.
- Achieved 2nd of 193 countries, scoring a record 172 points; featured on national news and TV.

Technical Skills

Languages: Python, C++, C, MATLAB, URScript, L^AT_EX, HTML, SQL, Assembly

Software: LT Spice, KiCad, SolidWorks, Onshape, Fusion 360, Ansys Discovery/Fluent, Cura, CorelDRAW, XFLR5, OpenRocket

Relevant Coursework

16-861: Mobile Robot Design (Space Robotics) – Terramechanics, Navigation, Sensing, Actuation

15-213: Computer Systems – Memory, Architecture, Concurrency, Systems Programming in C

18-100: Intro to ECE – Computer Architecture, Communication Protocols, Signal Processing